Wheel Alignment

Category: **Automotive** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Tyres and wheel alignment

What it is used for

A wheel alignment system measures wheel angles and helps adjust them to the vehicle manufacturer's specifications. Correct alignment improves vehicle stability, steering response, tyre life, and driving safety.

Main parts

- Wheel alignment lift or level alignment platform
- Wheel clamps and alignment targets or sensors
- Steering wheel holder
- Brake pedal depressor if required
- Alignment computer and vehicle specification database
- Turn plates and rear slip plates

Before you start

1. Confirm the vehicle is suitable for the alignment equipment and identify its correct model specifications.
2. Inspect the tyres for damage, uneven wear, correct pressure, and suitable tread condition.
3. Check wheel bearings, suspension joints, steering components, and ride height for defects.
4. Position the vehicle centrally on the level alignment platform and apply the parking brake as instructed.
5. Install wheel clamps and sensors or targets according to the equipment instructions.
6. Check that turn plates and slip plates are unlocked and that the alignment system is calibrated.

How to use

1. Select the correct vehicle make, model, year, and suspension configuration in the alignment system.
2. Compensate for wheel runout using the method specified by the equipment manufacturer.
3. Fit the steering wheel holder and brake pedal depressor if required by the procedure.
4. Follow the on-screen instructions to measure camber, caster, toe, and other supported angles.
5. Compare the measured values with the manufacturer's specification ranges.
6. Inspect adjustment points and identify worn or damaged parts before making adjustments.
7. Adjust the specified angles in the recommended sequence while keeping fasteners within approved torque values.
8. Recheck all readings and confirm that the values fall within the permitted ranges.
9. Tighten adjustment fasteners to specification and remove the measuring equipment carefully.

10. Complete the final inspection, perform a road test if authorized, and save or print the alignment report.

⚠ Safety

- Use the alignment lift only within its rated capacity and secure the vehicle at approved lifting points.
- Keep people clear of the vehicle while raising or lowering the lift.
- Do not perform alignment adjustments on damaged suspension or steering components without first addressing the defects.
- Keep cables, sensors, and tools clear of moving wheels and lift mechanisms.
- Use suitable protective equipment and follow workshop vehicle-movement procedures.
- Never rely on alignment readings alone to confirm that unsafe tyres or damaged steering parts are roadworthy.

Care & storage

- Clean wheel clamps, sensors, targets, and turn plates after use.
- Inspect cables, clamps, lift locks, and slip plates for damage or wear.
- Keep alignment software and vehicle specifications updated as authorized.
- Arrange calibration and preventive maintenance according to the equipment manufacturer's schedule.

Common mistakes to avoid

- Using incorrect vehicle specifications can produce unsuitable alignment settings.
- Ignoring tyre pressure or worn suspension parts can make measurements unreliable.
- Adjusting angles before checking for damaged components can hide underlying faults.
- Failing to secure fasteners or perform a final measurement can leave the vehicle incorrectly aligned.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.